

idTech4 Map Tools

Blender 4.x / 5.x .map Import / Export Add-on

Version 1.0.0 · Author: Samson & Claude Sonnet

1. Overview

The idTech4 Map Tools add-on imports whole levels from .map source files and exports worldspawn brush geometry back out again. It reads idTech4 brushDef3, Quake 3 ("standard") and Valve 220 three-point face formats, plus patchDef2 and patchDef3 Bezier patches, and it resolves and places the models the map's entities reference.

Supported operations:

- Import .map — brushes and patches as real Blender meshes, with materials and UVs
- Place the .ase / .lwo static models entities reference, at each entity's own origin and rotation
- Place MD5 skeletal models, each with its own armature and, optionally, its animation
- Place characters' heads and attachments — the models an entityDef declares through def_head and def_attach* — bone-parented to the joints of the body's skeleton
- Resolve entity classnames through the .def files' entityDef declarations, following inherit chains
- Apply .skin declarations, per placed entity
- Generate real, fully textured Blender materials for every face
- Export worldspawn brush geometry back to a .map as brushDef3, Version 2 or Version 3
- Validate Collection panel — check meshes against what a brushDef3 can actually hold, and jump straight to the offending faces

This add-on is one of six that make up the idTech4 Blender toolchain. It installs on its own, but static-model placement needs the companion idTech4 ASE / LWO Tools add-on, MD5 model and animation placement needs idTech4 MD5 Tools, and material generation needs idTech4 Materials. Each option is greyed out in the import dialog when its add-on is not installed and enabled, so nothing silently does nothing.

2. Installation

1. Open Blender and navigate to Edit > Preferences > Add-ons.
2. Click Install... and select idTech4_map_io.py.
3. Tick the checkbox next to "idTech4 .map Importer".
4. Close Preferences. The add-on is now active.

After installation the following entries appear:

- File > Import > idTech4 .map
- File > Export > idTech4 .map
- 3D Viewport > N-Panel > idTech4 > Sources
- 3D Viewport > N-Panel > idTech4 Map > Validate Collection

NOTE: Install the companion add-ons first if you want models and materials. The importer checks for them at run time, on every redraw of its own dialog, so enabling one in Preferences takes effect immediately without restarting Blender.

3. Source Directories

idTech 4 games do not ship their assets as loose files. Doom 3, Quake 4, Prey and The Dark Mod pack them into .pk4 archives; Doom 3 BFG packs them into .resources files. This add-on does not read inside either. It reads an ordinary directory tree on disk, and nothing else.

So before you set Base Directory, extract everything. A .pk4 is an ordinary zip archive — any zip tool will open one, and most will do it without renaming the file first. A .resources file is not a zip and needs a Doom 3 BFG extraction tool.

Extract every archive into one folder, preserving the paths stored inside them, so the result is the full idTech 4 directory structure:

```
<game>/
  base/                ← Base Directory points HERE
    def/               entityDef and model declarations
    dds/               precompressed textures
    guis/
    maps/              .map source files
    materials/         .mtr material declarations
    models/            .md5mesh, .md5anim, .ase, .lwo
    skins/             .skin declarations
    textures/          .tga / .png / .jpg source textures
    generated/         BFG only: .bimage, .bmd5mesh,
                      .bmd5anim, .base, .blwo caches
```

IMPORTANT: Base Directory is the "base" folder ITSELF — the one that CONTAINS

materials/, models/, textures/ and the rest — not the game folder above it, and not one of the folders inside it. Every path in a .mtr, .def, .skin or .map is written relative to this folder, so pointing one level too high or too low resolves nothing.

NOTE: A game ships several numbered archives (pak000.pk4, pak001.pk4, ... or game00.pk4, game01.pk4, ...) and the later ones patch the earlier ones. Extract them in ascending numeric order and let later files overwrite earlier ones — that is the same precedence the engine applies at run time. Extracting them in the wrong order silently leaves you with pre-patch assets.

For a mod or an expansion, extract its own archives into its own folder beside base — d3xp, darkmod, and so on — and set Mod Base Directory to that folder while leaving Base Directory on base. A mod supplies part of the tree and inherits the rest, and the add-ons search the two the same way the engine does.

NOTE: The extracted tree only has to contain what you actually intend to load. Sound, video and script archives can be left packed. The folders that matter are the ones in the diagram above.

Base Directory, Mod Base Directory and Materials Source live in a small JSON file under Blender's per-user config directory, shared by every idTech4 add-on and edited from the 3D Viewport N-Panel, idTech4 tab, Sources panel. The .map importer needs them for far more than materials: models, .def files and .skin files are all resolved under the same roots.

Setting	Required	Used for
Base Directory	Yes	models/, def/, skins/ and materials/ — everything the import resolves.
Mod Base Directory	No	An optional second tree searched BEFORE Base Directory, with Base Directory still searched behind it. This is the engine's own fs_game behaviour, and the fallback is per ITEM, not per tree: a mod shipping three .def files does not hide the base game's nine hundred.
Materials Source	No	One .mtr file, or a folder of them. Blank falls back to the "materials" subfolder under Mod Base (if set) and then Base Directory.

NOTE: The .map file itself normally lives in the extracted tree too, at <base>/maps/. Importing it from there is what lets "Derive from model" and the setup dialog's Derive Automatically choices work — both find the base folder by walking up from the .map file's own location. A .map kept somewhere else imports perfectly well, but you must set Base Directory by hand.

If anything ticked in the dialog needs a Base Directory and none is configured, the import stops and offers a one-off setup dialog once the file and every option are already chosen:

Choice	Effect
Select Now	Type or browse a Base Directory (and optionally a Materials path). "Set as default for future imports" is ticked by default and writes them to the shared config; untick it to use them for this import only.
Derive Automatically — just this time	Walk up from the .map file's own directory looking for a "materials" folder; Base Directory becomes that folder's parent. Nothing is saved.
Derive Automatically — and save settings	The same, but the derived paths are written to the shared config once they are actually known.
Skip — Don't Set Up Materials For This Import	Import geometry only, with blank placeholder materials.

4. Importing a Map

Use File > Import > idTech4 .map. A large map takes a while, so the import runs as a modal operation that keeps Blender responsive and reports progress as it goes. When it finishes, a full report opens in its own Text Editor window; the report is also left behind as a text datablock so it can be reopened later from any Text Editor.

4.1 Geometry Options

Option	Type	Default	Description
Scale Mode / Factor	<i>Enum / Float</i>	None / 1.0	None, the idTech4 ↔ Blender unit preset (inches to metres, ×0.0254), or a Custom Factor.
Rotation	<i>Enum</i>	0°	A pure Z rotation applied to everything imported: 0°, 90° (X to Y), -90° (Y to X), or 180°.
Worldspawn Geo Grouping	<i>Enum</i>	All	Worldspawn routinely owns thousands of brushes and patches. All combines them into a single mesh object; Patches combines only the patches; Brushes combines only the brushes; None gives every brush and patch its own object. All is dramatically faster to import and far lighter in the outliner.

4.2 Model Options

Option	Type	Default	Description
Import Static Models (.ase / .lwo)	<i>Bool</i>	On	Load and place the meshes referenced by entities' "model" key (func_static, misc_model, and so on), using each entity's

Option	Type	Default	Description
			origin and rotation. Needs the idTech4 ASE / LWO Tools add-on.
<code>Only import 1st layer of .lwo models</code>	<i>Bool</i>	On	Matches the engine, which only ever reads a .lwo's first layer. Has no effect on .ase models.
<code>Parse defs</code>	<i>Bool</i>	On	For entities with no explicit "model" key of their own, look up their classname in the .def files' entityDef declarations (following inherit chains) and import whichever model that entityDef declares — matching how idTech4 itself falls back to entityDef defaults for spawnargs the entity does not override.
<code>Shading</code>	<i>Enum</i>	Engine Shading	How the static models this import places are shaded. The same four choices the standalone .ase/.lwo importer offers; this dialog defaults to Engine Shading, which is how the level will actually look in game.
<code>Import MD5 Models</code>	<i>Bool</i>	On	Load and place MD5 skeletal models. Unlike .ase/.lwo, a .map entity never names an .md5mesh directly — it names a "model NAME { ... }" decl in a .def file, whose "mesh" line gives the actual path. Each placed entity gets its own Armature so each can carry its own pose, but instances the same underlying mesh data. Needs the idTech4 MD5 Tools add-on.
<code>MD5 Animations</code>	<i>Enum</i>	First Frame Only	What to do with the animation resolved for each placed model — see Section 4.3.
<code>Heads & Attachments</code>	<i>Bool</i>	On	Also place each character's head and whatever it carries or wears. An idTech4 character is several models, not one: the .map entity places only the body, while its entityDef's def_head and def_attach* keys name the head and each attached prop, each bound to a named joint of the body's skeleton with its own offset and rotation. Bone-parented, so they follow the body through an imported animation. Needs Import MD5 Models on.
<code>Apply Skins</code>	<i>Bool</i>	On	For entities with a "skin", swap specific materials on the model THAT entity places — only that instance. Checked in the same order the engine checks: the entity's own "skin" key, then its classname's entityDef skin (following inherit chains), then, for MD5 models, the "skin" line on the model decl the entity resolves to.

4.3 MD5 Animation Modes

How each placed model's animation is RESOLVED never changes — the entity's "anim" key, then the decl's idle / idleN / initial, then a numbered animN, then af_pose. Only what is DONE with the resolved animation differs.

Mode	Effect
First Frame Only (default)	Pose each model at the first frame of its animation and keep no Action. Models stand in their idle pose but do not move, and nothing about them is re-evaluated as the timeline plays. Measured on mars_city1: 22 fps in the viewport against 13 fps for Full, and a much smaller .blend.
Full	Bake the whole animation onto each model's Armature as an Action, so the models animate with the timeline. The faithful option, and by far the slowest to play back: every animated Armature makes Blender re-evaluate its pose and re-deform its meshes on every frame change.
None	Place the models with no animation at all, leaving each in its raw MD5 bind pose. No faster than First Frame Only and usually less accurate — most models' idle pose differs from their bind pose, so this leaves them visibly wrong.

4.4 Material Options

Option	Type	Default	Description
<code>Import Materials</code>	<i>Bool</i>	On	Build real, fully textured materials for every brush and patch face's material name, instead of blank placeholders. Needs the idTech4 Materials add-on.
<code>Material Mode</code>	<i>Enum</i>	Simple	Which of the Materials add-on's fidelity modes to build with: Maximum, Good, Basic or Simple. This dialog defaults to Simple where the model importers default to Maximum, because a whole level's worth of materials is a very different proposition from one model's.
<code>Parameters</code>	<i>Enum</i>	Baked	What to do with time, parm0..11, global0..7, sound and table lookups. Baked here, Dynamic in the model importers, for the same reason: on mars_city1, 219 drivers were 74 ms of a 91 ms frame.
<code>Derive from model</code>	<i>Bool</i>	Off	Ignore the paths below and derive them for this import by walking up from the .map file's own directory looking for a "materials" folder.
<code>Base / Mod Base / Materials</code>	<i>Display</i>	—	Read-only mirrors of what this import will use. Hover for the full path; edit them in the Sources panel.

IMPORTANT: Material Mode and Parameters are the two biggest levers on how usable the imported level is. Simple + Baked will open a large map on modest hardware; Maximum + Dynamic on the same map will not. Both can be raised afterwards for a subset of materials from the Materials panel.

4.5 What the Import Produces

One root collection named after the .map file, with one child collection per entity classname inside it:

```
Scene Collection
├─ mars_city1          ← named after the .map file
│   └─ worldspawn      ← the world brushes and patches
│       └─ func_static
│           └─ light
│               └─ monster_zombie_maint
│                   └─ ...
└─ ...                one collection per classname
```

Every entity's spawnargs are stamped onto its object as custom properties prefixed map_ — map_classname, map_origin, map_name and so on — so nothing from the source file is lost, and the exporter can write worldspawn's own keys back out again.

The import report lists, separately: brushes imported and skipped, patches imported and skipped, models imported and failed with a static/MD5 breakdown and per-format timings, heads and attachments placed, how many head joints were driven from the body, particle definitions encountered (these are counted, not treated as failures — they are never a mesh this importer could load), and the full materials build summary.

NOTE: The import builds outside the view layer and reveals everything at the end. This is what makes a large map import in a reasonable time — many Blender calls get more expensive as the view layer fills up — and it is why nothing appears in the viewport until the run completes.

5. Exporting a Map

Use File > Export > idTech4 .map. The exporter writes worldspawn brush geometry as brushDef3 — the exact inverse of the brushDef3 half of the importer, down to the engine's own float formatting, so a re-export of an imported map diffs against the original on the brushes that changed and nothing else.

IMPORTANT: Only worldspawn brush geometry is exported. Patches, entities, lights and placed models are not written back.

5.1 Source

Option	Type	Default	Description
Source	<i>Enum</i>	Worldspawn Collection	Worldspawn Collection: walk up from the active object to its root collection, then export the meshes in that root's "worldspawn" child — the layout an import produces. It refuses, naming what it did find, when there is no such collection. Active Collection: export every mesh in the collection highlighted in the outliner and its children, for scenes built by hand. Selected Objects: only the selected meshes.
Skip Hidden Objects	<i>Bool</i>	On	Leave out anything whose outliner EYE is closed.
Apply Modifiers	<i>Bool</i>	Off	Export each object's modifier-evaluated mesh instead of its raw geometry. Off by default so the mesh in the outliner is the mesh that gets written.
Keep Worldspawn Spawnargs	<i>Bool</i>	On	Write back the key/values the worldspawn entity came in with (call, script, and anything else the import stamped on as map_*). Turn it off to write a bare worldspawn.

5.2 Transforms

The same Transforms box as everywhere else: Scale Mode (None, the Blender → idTech4 metres-to-inches preset, or a Custom Factor) and a Rotation preset applied as a Z rotation on the way out.

NOTE: Origin-relative sums lose the sign about 8000 units out from the origin, which is simply where float32 resolution runs out — not an exporter defect. Judge a round trip against that floor rather than against exact equality.

5.3 Map Format

Option	Type	Default	Description
Version	<i>Enum</i>	Version 2	Version 2 (Doom 3 / Prey / Dark Mod) writes "Version 2" and ends every brush side with the trailing "0 0 0", exactly as idMapBrush::Write and these games' editors do. Version 3 (Quake 4) writes "Version 3" and omits it, matching Quake 4's own shipped maps.

Option	Type	Default	Description
Snapping	<i>Enum</i>	Normals Only	How exported planes are tidied up — see the table below.
Grid Size	<i>Float</i>	1.0	The grid vertex positions are rounded to, in idTech4 game units, before any plane is derived. Only shown for the two grid snapping modes.

Snapping mode	Effect
Normals Only (default)	Snap a plane normal to an exact axis when it is already within 1e-5 of one, then re-derive the plane distance from the snapped normal. Vertex positions are never moved, so this cannot break geometry — it only stops an axis-aligned brush being written as "(0.00000004 0 1 -63.9999992)".
Normals + Grid	Normal snapping as above, and additionally round every vertex position to the grid size before any plane is derived.
Grid Only	Round every vertex position to the grid and write plane normals exactly as they come out.
None	Write the transformed values exactly. Most faithful to the scene; axis-aligned brushes may read as very slightly rotated when reopened in a level editor.

IMPORTANT: Moving vertices can make a face that was flat non-flat, or a brush that was convex concave. Both grid modes can therefore break geometry — anything the snap breaks is reported and refused rather than written.

5.4 Geometry Check and Report

The export dialog carries a Check Geometry button that runs the same checks the export itself runs, with the dialog's own Source, Skip Hidden and Snapping settings, and summarises the result in place with a count per issue type. The full, clickable detail appears in the idTech4 Map sidebar tab (Section 6).

Option	Type	Default	Description
Open Report	<i>Bool</i>	On	Open the full export report in a Text Editor window afterwards. The report is written to a text datablock either way and can be reopened from any Text Editor.

The export ends on an INFO, WARNING or ERROR status depending on what it found, and an ERROR-level result cancels rather than writing a file. Either way the sidebar's issue list is filled from the export's own findings, so Highlight Issue works on exactly what was just rejected without having to run the check again.

6. The Validate Collection Panel

3D Viewport > N-Panel > idTech4 Map tab. It checks every mesh for the four things a brushDef3 has to be: closed, flat-faced, convex, and one material and one flat UV projection per plane.

6.1 Why These Four

A brushDef3 is an intersection of half-spaces, so the only mesh that survives the round trip is a closed convex polyhedron. Each edge-connected island is checked for exactly that — closed manifold, every face flat, every vertex behind every face plane, four or more distinct planes — which is not merely necessary but sufficient: an island that passes re-imports as the same solid.

Coplanar faces merge onto one side, so a triangulated box is still six sides. Coplanar faces carrying different materials are refused, since a side has room for exactly one. Texture matrices are RECOVERED rather than approximated: three non-collinear loops determine the 2×3 matrix exactly, and the residual is then measured over every loop of the side, so a face whose UVs are not one flat projection — an unwrapped island, a packed atlas, two coplanar faces with unrelated UVs — is refused rather than least-squares-fitted. Approximating it would silently reshuffle the texturing instead of saying so.

6.2 Issues Reported

Issue	Severity
Non-flat face	Error
Degenerate face	Error
Open or non-manifold edge	Error
Concave brush	Error
Not a solid (under 4 sides)	Error
Mirrored plane pair	Error
Coplanar faces, 2+ materials	Error
UVs are not a flat projection	Error
Over the dmap brush-side limit	Error
Over the point limit on one side	Error
Outside the world bounds	Error
No material assigned	Warning
No UV map	Warning
Normals face inward	Warning

Errors stop a brush being written; warnings do not. The list groups by issue type with the fatal kinds first, and each row names the object, the brush index within it and how many mesh elements are affected.

6.3 Highlight Issue

Select a row and press Highlight Issue: the add-on selects the exact faces, edges or vertices responsible, in the right select mode, on the right object. If the mesh has been edited since the check was run, the recorded indices no longer address the same elements, so the button refuses rather than selecting whatever now happens to live at those numbers — run the check again.

NOTE: The issue list is stored on the Scene, so it survives a save. You can validate a large map, close Blender, and pick the list up again later.

7. Format Notes

7.1 Face Formats Read

Format	Notes
brushDef3	The native idTech4 form: a plane (normal + distance) and a 2×3 texture matrix per side. This is the only form the exporter writes.
Quake 3 "standard"	Three points defining the plane, with shift/scale/rotate texture parameters.
Valve 220	Three points plus explicit U and V axes. The explicit axes already produce Blender-compatible orientation, so V is not flipped for these.
patchDef2 / patchDef3	Bezier patch surfaces. patchDef3 carries explicit subdivision counts; patchDef2 is subdivided by the importer.

7.2 Decl Lookup

entityDef and model decl names are matched the way the engine matches them: backslashes converted to slashes, lowercased, and truncated at the last dot. Decl keywords themselves are case-insensitive — 268 Dark Mod declarations spell it "entitydef" — so a case-sensitive scan would drop them silently and the entity would resolve to nothing.

Inheritance is prefix-matched: inherit, inherit1, inherit2 and inherits all inherit, and file order decides which wins.

7.3 Heads and Attachments

Head and attachment binding rules differ per game, and the Blender constraints that drive a head's joints from the body's are only created when the two rigs actually agree on that joint's rest pose. Where they do not, the joint is left alone and the import report says how many were skipped for that reason.

7.4 Skins

A skin makes an object's material slots and its mesh's own materials diverge, because the swap is per placed instance. Both have to be scanned when collecting the material names a map needs; scanning only one of them leaves the other set rendering flat white.

8. Tips and Troubleshooting

Import Is Slow

Set Worldspawn Geo Grouping to All, Material Mode to Simple and Parameters to Baked, and leave MD5 Animations on First Frame Only. Full MD5 animation is the single most expensive option in the dialog, followed by Maximum materials with Dynamic parameters.

An Option Is Greyed Out

Its companion add-on is not installed or not enabled. The dialog says which one, in red, directly under the greyed row.

Nothing Resolves At All

If every lookup fails — no materials found, no models found, no .def files found — the usual cause is that the game's assets are still inside their .pk4 or .resources archives. These add-ons cannot read into an archive; everything has to be extracted to a real directory tree first. The second most common cause is Base Directory pointing at the game folder rather than at the base folder inside it: it must be the folder that directly contains materials/, models/ and textures/.

Everything Is Pink or Flat White

The material decls or their textures could not be found. Check Base Directory, and read the import report's materials section — it names every material that fell back to a placeholder and why. For The Dark Mod, note that around 99% of textures exist only under dds/, so a resolver pointed at a tree without that folder will find almost nothing.

A Quake 4 Skybox Looks Wrong

It probably is not. Quake 4 skyboxes are small detached sky rooms placed beside the level rather than shells around it, so seeing one off to the side of the map is correct content, not an import fault.

Export Refuses "No worldspawn collection"

With Worldspawn Collection source, the exporter walks up from the active object to its root collection and looks for a child named worldspawn. Either select something inside an imported map first, or switch Source to Active Collection.

Brushes Are Rejected on Export

Run the check and read the reasons — they are specific. Note also that some shipped maps contain brushes carrying sides Blender cannot represent at all; those come in already unable to round-trip, so subtract them before concluding the exporter has drifted.

Reopening the Reports

Both the import and export reports are ordinary text datablocks. Open any Text Editor and pick them from the datablock browser — they are named after the file they describe.

idTech4 Map Tools — end of documentation